

# Prediction of Corona Virus Outbreak using Machine Learning

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**Abstract**—Predicting the corona virus can be divided into several phases, including a state-wide analysis that includes active, confirmed, cured, deaths as well as an increase in cases on a daily basis that includes each and every state of India as well as Union Territories. This also includes a thread of new corona virus cases from throughout India and forecasts the outbreak's conclusion in the next days. Machine learning algorithms like SVM, Linear Regression and Decision Tree Regression are used to analyze this data and improve this model's outcome. In this study, Jupyter notebook is used which provides an environment that is suited for machine learning principles. This technique provides for a comprehensive analysis of the virus's spread, including total and active cases, as well as forecasting future outbreaks and a weekly study epidemic.

**Keywords**—outbreak, regression, visualization

## I. INTRODUCTION

Corona virus epidemics impact the whole human population and can result in catastrophic disease and death. Corona viruses are a big viral family that may infect people in a number of ways. The first documented endemic of this virus family was Severe Acute Respiratory Syndrome (SARS), which occurred in 2003. However, the second severe illness epidemic occurred in 2012 in Saudi Arabia with Middle East Respiratory Syndrome (MERS) which is a condition that affects people in the Middle East. The current corona virus outbreak is widespread and it was first detected in late December 2019 in China particularly in the state of Wuhan state [1]. Three medical students from India who had returned from Wuhan, first reported COVID-19 in India, in three different towns of Kerala. Considering the situation, Kerala was put on lockdown on March 23, 2020, followed by rest of the country on March 25, 2020. By respiratory droplet, Covid-19 spreads quickly from one person to another [12]. Corona virus spreads mostly by droplets produced by infected people during coughing, sneezing, and direct contact with infected people. These can occur as a result of direct or indirect interaction with an infected individual. To prevent the spread of COVID-19, experts advise that individuals should avoid gatherings, stay at home, avoid physical contact, keep a safe distance of at least two meters between them, wash their hands often with water and soap, and isolate themselves if any signs of sickness appear. The disease causes respiratory diseases such as influenza, flu, and Ebola, with symptoms such as fever, dry cough, breathing difficulties, and diarrhea [1] [13]. In 2-14 days, those infected with the corona virus begin to show symptoms [14].

The World Health Organization spent a great deal of time and effort identifying the most frequent misconceptions that might have a negative impact on community health. The most fascinating aspect of its spread is that it may be carried for several days without generating symptoms. The WHO offers coronavirus guidance on its website and through different social media platforms such as Twitter, Facebook, Instagram, and others. Finally, the corona virus pandemic is a huge concern, because learning has an impact on the most fundamental aspects of life. It devastates the political, social, economic, religious, and financial institutions of the globe.

The most popular application of artificial intelligence is Machine learning (ML) which allows a system to learn and develop itself without involvement of any software [1]. Check the patterns in the data and adjust the behavior of the program accordingly [2]. The healthcare industry uses machine learning techniques such as Support Vector Machine, Linear Regression, Polynomial Regression, and others to predict, identify, and analyze outbreaks such as the corona virus. For predicting the transmission of an epidemic, several studies use machine learning based upon statistics or to use fresh data to evaluate the parameters of epidemiological models. Artificial intelligence (AI) and machine learning (ML) approaches play a key role in recent research in reducing the effects of virus spread [11] Different approaches to using machine learning (ML) in patient data are explored [7].

Machine learning is an essential component of data science, which is continuously expanding. Statistical techniques are used to train algorithms to provide predictions. We can successfully anticipate future outbreaks and take steps or place the region under lockdown to minimize active cases by researching the cases.

Corona virus outbreak prediction can be performed by utilizing the datasets gathered from various resources. Taking safeguards in this situation, keeping the rising number of active cases under control and the growing number of instances makes it difficult to keep track of and treat each individual.

In this work, we did a basic analysis on a Kaggle dataset and used a model called SVM to predict the outcome. The dataset provides the most recent COVID-19 state-by-state statistics in India up to July 07, 2021. Machine learning might be used to compare total cases, active cases, recovered patients, and deaths in Indian states/UTs due to the corona virus. The

proportion of people who have been rescued and the percentage of people who have died in Indian states/UTs are examined. The state-by-state research may be used to forecast how the outbreak will play out in the future. We will be able to determine which state performs best in terms of medical facilities when compared to the other states by finishing this survey.

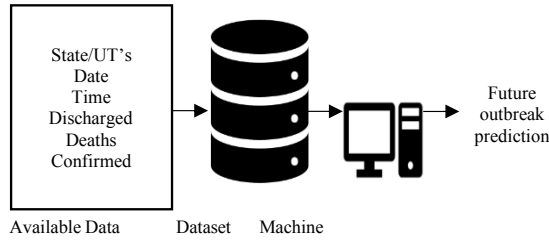


Fig. 1. System architecture of prediction of corona virus outbreak

## II. RELATED WORKS

Various artificial intelligence and machine learning algorithms were used to assess and forecast Covid-19 disease by different researchers. Several epidemiological models have been widely investigated in the literature in recent decades, presenting influence diffusion as a sickness that spreads at a certain rate over biological groups. One of the most commonly utilised models is the Susceptible-Infected-Recovered (SIR) model, in which a person develops permanent immunity or is removed after recovering from a disease.

A machine learning method to coronavirus illness prediction was proposed by R. Sujath, *et al.* [1]. Throughout their study, they presented a useful model for anticipating COVID-19 spread. They used a variety of machine learning methods, including linear regression, multilayer perceptron, and neural networks. The best predictors were found to be vector auto regression and a multilayer machine-learning method.

F. Amira *et al.* [5] offer a COVID-19 epidemic data analysis and corona truck forecasting. They have proven the importance of sharing information that can assist enhance reaction time and reduce the danger of COVID-19 throughout their study. The existing state of affairs was discovered via the research. The corona tracker displays the pattern of coronavirus illness, which may also be anticipated using SEIR analytics.

Giordano and Giordano *et al.* [3] in Italy, the basic SIR model was enhanced to include diagnosed/undiagnosed cases, symptom severity, and the difference between recovered and died patients. They model a dynamic system with five states and three outputs, with 16 parameters that are changed experimentally to account for the various levels of social-distancing remedies adopted by the Italian government.

F. Ardabili *et al.* [6] studied a multilayer perceptron, adaptive network-based, fuzzy inference technique and built a machine-learning model to evaluate and anticipate the COVID-19 epidemic. Based on their research, they presented machine learning as well as methodologies for replicating the

COVID-19 pandemic. The outcomes of two Machine Learning models after the experiment.

Using a maximum-hunting parameter estimate technique and an enhanced susceptible exposed infection recovered (SEIR) model, Z. Zhao *et al.* [9] forecast COVID-19 in various African nations. The most practicable updated model, which differs from the usual SEIR models, was chosen. The model was thought to be a better representation of the latest quarantine status and intervention measures.

F. Rustam *et al.* [8] employed a machine-learning system to anticipate probable coronavirus patients to aid decision-making. They utilised four common prediction methods, including linear regression, list absolute shrinkage, vector support, and exponential smoothing, in their study. The number of new incidents was predicted using models.

Yan *et al.* [4] calculated the criticality and survival probability of patients with severe COVID-19 infection using numerous risk factors and demographic data. Out of over 300 clinical features, they used an XGBoost (XGB) model to narrow down the list to only three: lactic dehydrogenase (LDH), lymphocyte, and high-sensitivity C-reactive protein (Hs-CRP). The proposed model was validated using data from 29 individuals. The model's capacity to evaluate mortality risk with 0.95 precision and 0.90 prediction accuracy was one of the study's major findings.

Sun *et al.* [7] used a support vector machine (SVM) prediction model to estimate the severity of COVID-19 patients. The investigation used clinical and laboratory variables that are significantly connected to these individuals. They determined that age, growth hormone secretagogues (GHSs), immunological feature cluster of differentiation 3 (CD3) percent, and total protein are the essential features that identify moderate and severe cases using 336 COVID-19 individuals, 26 severe/critical cases, and 310 noncritical cases.

An *et al.* [10] used multiple machine learning techniques, including LASSO, SVM (linear and RBF), RF, and KNN, to create a model to predict COVID-19 patient mortality. Within 14 and 30 days following the initial diagnosis, the models were trained to detect two cases: death and survival. With an accuracy of 0.962, sensitivity of 0.92, and specificity of 0.91, the competition was outperformed by Linear SVM. The researchers discovered that age, diabetes, and cancer were all important factors in COVID-19 patients' death predictions.

## III. MODEL PROPOSED

Dataset extraction is the concept used by ML [8]. To forecast how the model will be used, ML algorithms are trained on labelled and unlabelled train datasets. There are three different types of machine learning. Machine learning may be divided into three categories namely supervised, unsupervised, and reinforcement. There is an input variable X and an output variable Y in supervised ML, and it utilises an algorithm to learn the mapping function from the input to the output, which is  $Y=f(X)$ . The objective of mapping function input X to output Y [8] is to predict the outcome.

Our suggested system is a model that forecasts the virus's spread over the following few days. In the research of corona virus, a strong model that predicts how the virus could spread

across different states is required. The normal models and statistical models are receiving more attention than the long-term prediction models. This is because of the difficulty in finding the essential data required for the long-term prediction models. In order to make this simple, the existing models need to be upgraded. The existing models show the technology use in studying, analyzing and predicting the corona virus.

| Sno | Date | Time       | State/UnionTerritory | ConfirmedIndianNational | ConfirmedForeignNational | Cured | Deaths | Confirmed |
|-----|------|------------|----------------------|-------------------------|--------------------------|-------|--------|-----------|
| 0   | 1    | 30-01-2020 | 06:00 PM             | Kerala                  | 1                        | 0     | 0      | 0         |
| 1   | 2    | 31-01-2020 | 06:00 PM             | Kerala                  | 1                        | 0     | 0      | 0         |
| 2   | 3    | 01-02-2020 | 06:00 PM             | Kerala                  | 2                        | 0     | 0      | 0         |
| 3   | 4    | 02-02-2020 | 06:00 PM             | Kerala                  | 3                        | 0     | 0      | 0         |
| 4   | 5    | 03-02-2020 | 06:00 PM             | Kerala                  | 3                        | 0     | 0      | 0         |

Fig. 2. Dataset (First 5 values are shown)

The model utilizes accessible data from datasets such as States/UT's, total cases, discharged, and deaths. From the datasets, the model is validated by training it, and then the hyper parameters are applied using Support Vector Machine, Linear Regression, Decision Tree Regression and some other methods.

A heat map is used to determine the relationship between each and every feature. The correlation matrix is used in the heat map. The results are visualized using graphs.

#### IV. PROCESS

Collecting, cleaning, choosing a model, training and visualizing are the major steps in this process. In this study, we collected the data from a popular resource called Kaggle. We cleaned the collected data by removing some unnecessary parameters like ConfirmedIndianNational and ConfirmedForeignNational and checked for data sufficiency. The collected data is stored in the system and we train the



Fig. 3. Outbreak prediction visualizing process

machine using suitable algorithms. We assigned 20% of our data for testing and 80% for training. Based on our algorithm provided, the machine generated our required output. The accuracy of a model is determined by the quantity and quality of the data. The end result of this phase is usually a data representation that is utilized for training. Using pre-collected data from Kaggle, UCI, and other sources still counts as part of this phase.

Some of the useful algorithms that is best suits for this kind of projects are Support Vector Machine, Regression, Neural Networks and Decision Tree Regression. Some of them which are used in this study are explained below:

##### A. Data Visualization

Data visualisation in machine learning is crucial for understanding how data is utilised in a machine learning model and for assessing it. Facets is a free, open-source Python module that may be used to quickly display and

analyse data. Facets is made up of two separate visualisations: one for comprehending the data and the other for studying it.

The display of data using conventional visuals such as charts, plots, infographics, and even animations is known as data visualization. These easy-to-understand visual representations of data communicate complicated data linkages and data-driven insights.

##### B. Support Vector Machine

SVM (Support Vector Machine) is a supervised machine learning (ML) classification and regression method [8]. In today's world, SVM is highly popular and generates an effective prediction algorithm. SVM is built somewhat differently than other machine learning algorithms, and it can do classification, regression, and out layer prediction. A distinct hyperplane is used to officially build SVM, which is a discriminative classifier. It is a representation of instances as points in space that are mapped with a gap as large as feasible between the points of various categories. The linear SVM is the most basic, with the hyperplane located in the space of the input data  $x$ .

##### C. Regression

Regression analysis is a collection of machine learning algorithms for predicting a continuous outcome variable ( $y$ ) based on the values of one or more predictor variables ( $x$ ). In a nutshell, the purpose of a regression model is to create a mathematical equation that specifies  $y$  as a function of  $x$  variables.

Observations from the data are used to forecast continuous values. The link between the dependent variable and the independent variables can be determined using a statistical approach called regression.

#### V. ANALYSING AND VISUALISING OF DATA

From the dataset that we took, we performed some analysis and produced graphical representation of data that tells us about how the effect of virus is being produced across various states in India. This analysis gives the accurate information of past situation and is useful for the approximate prediction of future situation. For this, we used Jupyter Notebook which provides a comfortable user interface and environment that supports machine learning principles.

We performed various operations on the dataset that we collected from a well-known resource called as Kaggle. The main advantage of this analysis is that we can know which state is severely affected with the virus and which state has the greatest number of active cases. Based on this we can apply different strategies in different regions in order to take situation under control.

Analysing and visualising the data is most important in terms of understanding the overview of any project. The best part of performing this analysis is that it provides a graphical representation of data that is flexible for fetching particular information and easy to understand the overall view of the situation across the country.

#### VI. RESULT

In this study, we first performed the analysis of available data which consists of details about COVID-19 spread across

India previous to July07, 2021 and applied ML models like SVM, Linear Regression and Decision Tree Regression in

order to forecast the future outbreak. The results that are obtained from our study are:

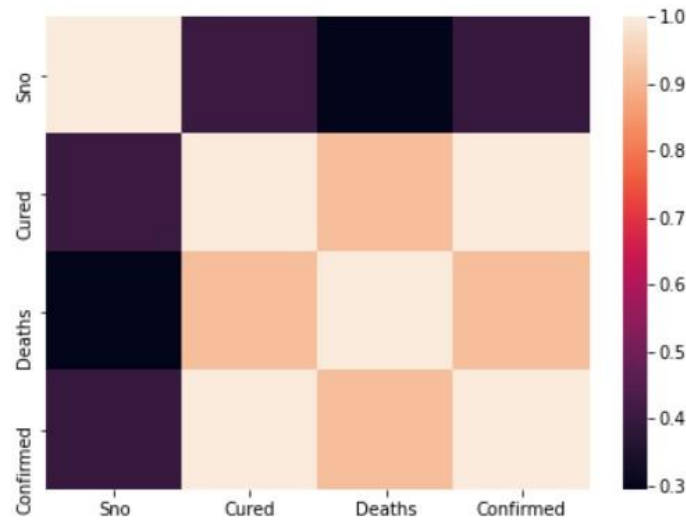


Fig. 4. Correlation between every attribute of the dataset

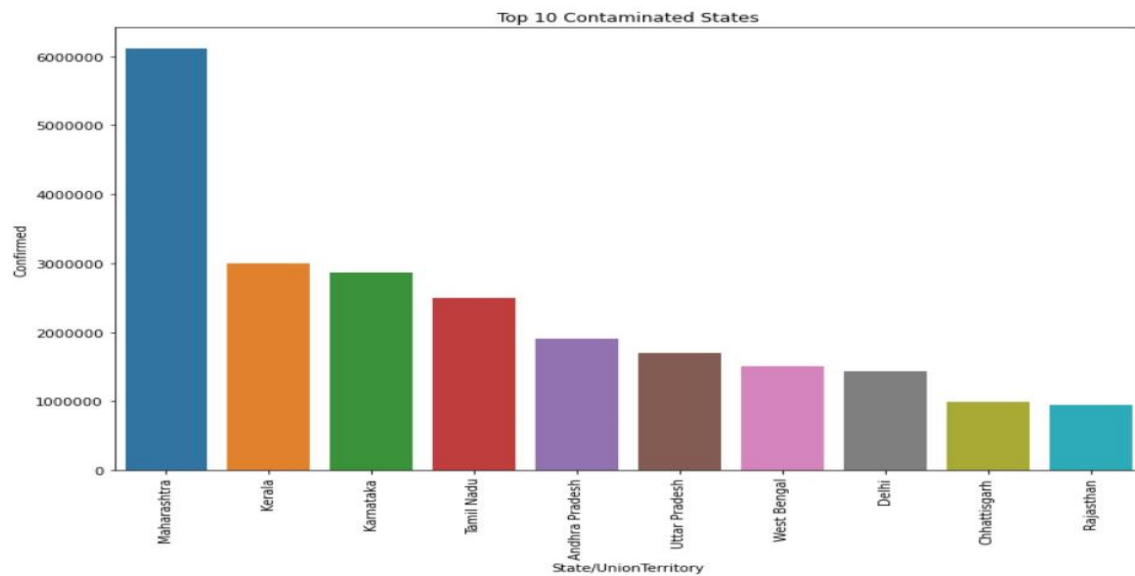


Fig. 5. Top 10 contaminated states

Fig.3 shows us the correlation between the variables. When one variable's value changes, the other tends to change in the same direction. The strength and direction of this tendency to vary together are measured by a correlation coefficient. There is a linear relationship between confirmed cases and cured cases. This implies that number of cured cases corresponds to the increasing number of confirmed

cases. That means the recovery rate is also more corresponding to the newly occurring cases. If we observe that the relation between deaths and cured cases, we found that 92.4% of people who admitted to the hospital are getting

recovered and discharged. However, the death percentage stays as 7.6 which is not much at this point of time but may increase severely.

Fig.4 shows us the top 10 contaminated states across India. We observed that Maharashtra has maximum number of cases followed by Kerala, Karnataka and Tamil Nadu whereas Mizoram, Sikkim and Nagaland were less effected with this pandemic. On the other hand, Union Territories like Andaman and Nicobar Islands, Lakshadweep, Dadra and Nagar Haveli and Daman and Diu are not much affected with this virus. From this analysis, we came to know that Maharashtra should implement strict safety rules and states

like Mizoram have to be implemented with precautionary measures.

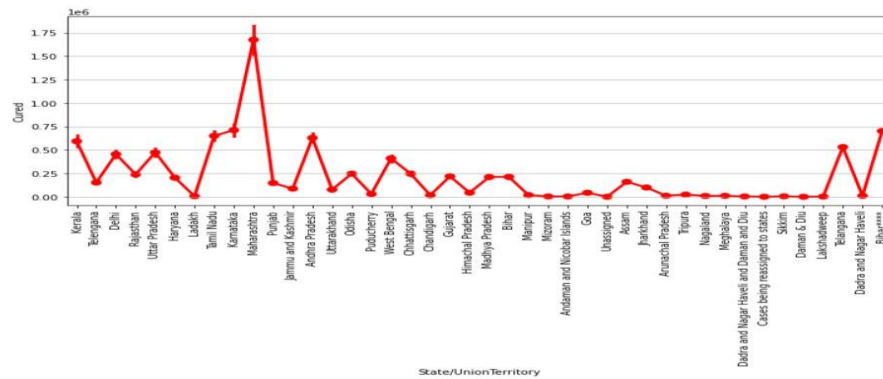


Fig. 6. No. of cured cases state wise

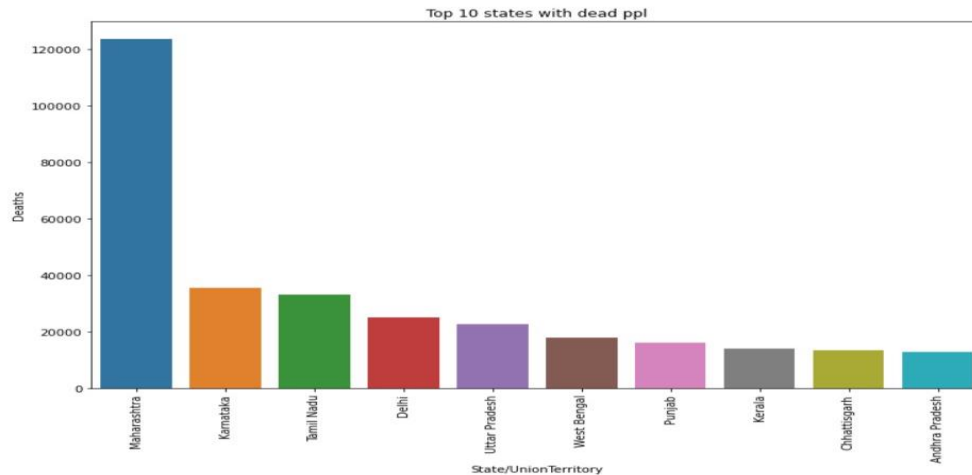


Fig. 7. Top 10 states in terms of deaths

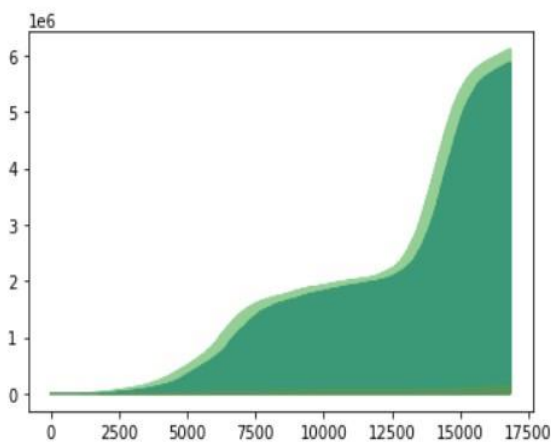


Fig. 8. Overview of cases

Fig. 5 shows that Maharashtra is pretty high in terms of number of cured cases because it is the one with highest

number of cases as well. Fig. 3 shows us the relationship between confirmed cases and cured cases somewhat strong and that's how Maharashtra is good at recovery rate and Fig. 7 shows us the overview of COVID-19 cases across India.

On analysing these statistics, we found an interesting factor to note from our paper that is despite having highest total number of cases and active cases Maharashtra does not have the highest death ratio and surprisingly Punjab has the highest death ratio despite having a smaller number of total cases on comparing with the other states. On the other hand, Mizoram is the state with least death rate. Among Union Territories, Daman and Diu is the one with strong control over the situation. The number of cases is approximately proportional to the number of people who have been discharged.

## VII. CONCLUSION AND FUTURE ENHANCEMENTS

The COVID-19 epidemic has wreaked havoc throughout the globe, prompting a countrywide health emergency. To counteract the epidemic, several measures have been made.

While analyzing our outputs from this project, we discovered that Maharashtra has the greatest number of total cases and active cases. However, the same state is doing well in terms of providing medical services.

We also found that Punjab need more attention in terms of medical facilities, based upon the high death ratio when compared to the other states.

While a few states require more care in terms of preventing COVID-19 contamination by various preventive measures including lockdown if necessary. By using this analysis, we predicted how cases varies in number in different states of India in future. From our proposed models like SVM, Linear regression and Decision Tree Regression, Linear Regression shown the better outcome.

In this project we performed three machine learning models which gives us approximate prediction of future based upon the details of past. However, this data can be further used to improve the model's outcome. With certain models like Neural networks, Polynomial regression and many other models which has already been implemented by many researchers, we can find out which is doing better.

```
# Decision Tree
from sklearn.metrics import r2_score #Checking for accuracy
prediction1=tree.predict(x_test)
score1=r2_score(y_test,prediction1)
score1

0.9998591938654751
```

Fig. 9. Accuracy shown by Decision Tree Algorithm

```
# SVM
from sklearn.metrics import r2_score
prediction1=model.predict(x_test)
score3=r2_score(y_test,prediction1)
score3

0.9998770191223476
```

Fig. 10. Accuracy shown by SVM Algorithm

So far, many researchers provided many models on the same research. Out of those models the best model for every research comes into picture. For instance, many projects revealed that Vector Auto Regression and Multilayer machine learning algorithms were chosen as the best predictors. SIR model is the best approach to predict the severity of the virus in once body. To determine the quarantine status and intervention measures SEIR (Susceptible Exposed Infection Recovered) model is chosen to do better compare to the others. In order to estimate the criticality and survival chances of patients with severe COVID-19 infection, XGBoost (XGB) produces the better outcome but Linear SVM (Support Vector Machine) performs better in this case. All these researches not only useful for predictions but they are very helpful for doctors in order to produce the best medicine to the patients based on the severity of virus and governments to take necessary precautions to control the spread of virus in vast amount.

Future work will focus on adding more data to our analysis and this time we want to examine how this data may enhance epidemiological model predictions. We also want to look at customizing the SVM model which is very efficient model in predicting the future outbreak and detecting the severity and criticality so that we can better mimic the characteristics of

epidemic spread. Finally, our aim is to build a model that detects what variant of virus is infected into the body using machine learning algorithms.

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